Cheol Hyun Park

Ph.D.

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Summary

Nanomaterials, self-assembly, ceramics, biomaterials, and fracture mechanics.

Education

MS/PhD combined — **Sungkyunkwan University**, Materials Science and Engineering — Suwon, ROK — Jan 2020 – Jan 2026
- Nanoparticle Self-Assembly, Ceramics, Biomaterials, Fracture Mechanics

BS — **Sungkyunkwan University**, Materials Science and Engineering — Suwon, ROK — Jan 2014 – Jan 2020

Experience

Institute for Advanced Study, Princeton University, Professor of Theoretical Physics — Princeton University, NJ — Jan 1933 – Jan 1955 — 22 years 1 month

Teaching at Palmer Physical Laboratory (now 302 Frist Campus Center). While not a professor at Princeton, I associated with the physics professors and continued to give lectures on campus.
- Relativity
- Description 2.

Volunteer

People's Climate March, Lead Organizer — Zurich, Switzerland — Apr 2014 – July 2015

Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.
- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

Awards

Nobel Prize in Physics — Nov 1921

The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'
Royal Swedish Academy of Sciences
www.nobelprize.org/prizes/physics/1921/einstein/biographical

Publications

Bioinspired adenosine triphosphate as an "all-in-one" green flame retardant via extremely intumescent char formation — Jan 2021

Jeong, Sun Hwan and Heo, Jun Hyuk and Lee, Jin Woong and Kim, Min Jeong and Park, Cheol Hyun and Lee, Jung Heon
doi.org/10.1021/acsami.1c02021 (American Chemical Society)

Synergistic adenosine triphosphate/chitosan bio-coatings on polyurethane foam for simultaneously improved flame retardancy and smoke suppression Jan 2023
Song, Hyewon and Park, Cheol Hyun and Jeong, Sun Hwan and Heo, Jun Hyuk and Lee, Jung Heon
doi.org/10.1021/acsapm.3c00514 (American Chemical Society)

Nanoparticles as next-generation tooth-whitening agents: Progress and perspectives Jan 2022
Kim, Dai-Hwan and Bae, Jina and Heo, Jun Hyuk and Park, Cheol Hyun and Kim, Eun Bi and Lee, Jung Heon
doi.org/10.1021/acsnano.2c01412 (ACS Nano)

Natural bone-mimicking nanopore-incorporated hydroxyapatite scaffolds for enhanced bone tissue regeneration Jan 2022
Kim, Chansong and Lee, Jin Woong and Heo, Jun Hyuk and Park, Cheolhyun and Kim, Dai-Hwan and Yi, Gyu Sung and Kang, Ho Chang and Jung, Hyun Suk and Shin, Hyunjung and Lee, Jung Heon
doi.org/10.1016/j.biomaterials.2022.109792 (Biomaterials Research)

Biomolecules as green flame retardants: recent progress, challenges, and opportunities Jan 2022
Jeong, Sun Hwan and Park, Cheol Hyun and Song, Hyewon and Heo, Jun Hyuk and Lee, Jung Heon
doi.org/10.1016/j.jclepro.2022.133241 (Journal of Cleaner Production)

Progress and perspectives of metal-ion-substituted hydroxyapatite for bone tissue engineering: comparison with hydroxyapatite Jan 2022
Kim, Si Hyun and Park, Cheol Hyun and Heo, Jun Hyuk and Lee, Jung Heon
doi.org/10.1166/jkscs.2022.6321 (Journal of the Korean Ceramic Society)

A multifaceted heat-switching nanofiller: Design of heat dissipation–insulation switching materials above a critical temperature Jan 2024
Song, Hyewon and Jeong, Sun Hwan and Park, Cheol Hyun and Kim, Min Jeong and Kim, Hyunlee and Heo, Jun Hyuk and Lee, Jung Heon
doi.org/10.1016/j.cej.2024.154406 (Chemical Engineering Journal)

Monolithic DNApatite: An Elastic Apatite with Sub-Nanometer Scale Organo–Inorganic Structures Jan 2024
Lee, Jin Woong and Lee, Byoungsang and Park, Cheol Hyun and Heo, Jun Hyuk and Lee, Tae Yoon and Lee, Dongtak and Bae, Jina and Sundharbaabu, Priyannth Ramasami and Yeom, Won Kyun and Chae, Sudong and others
doi.org/10.1002/advs.202406251 (Advanced Materials)

Simultaneous dyeing, flame-retardant, and antibacterial treatment of fabrics using adenosine triphosphate as a multifunctional agent Jan 2025
Park, Cheol Hyun and Song, Hyewon and Kim, Dai-Hwan and Kim, Junseok and Jeong, Sun Hwan and Rayappan, John Bosco Balaguru and Heo, Jun Hyuk and Lee, Jung Heon
doi.org/10.1016/j.cej.2025.159912 (Chemical Engineering Journal)

Transformative Mechanical and Piezoelectric Properties in Organic–Inorganic Hybrid Materials for Biomedical Applications Jan 2026
Cho, Jaewon and Park, Cheol Hyun and Ramasami Sundharbaabu, Priyannth and Ju, Seong-Uk and Shin, Davin and Heo, Jun Hyuk and Kim, Miso and Lee, Jung Heon
doi.org/10.1021/acsnano.5b08724 (ACS Nano)

Skills

Materials Characterization: Electron Microscopy, Indentation method, Universal tensile machine, X-ray diffraction, X-ray fluorescence, X-ray absorption spectroscopy

Biological assays & cell culture: Cell culture (MC3T3-E1, L929), cytotoxicity assay, cell proliferation assay, fluorescence microscopy

Languages

Korean: Native speaker

English: Fluent

Interests

Materials Science and Engineering: Nanomaterials, Self-assembly, Ceramics, Biomaterials, Fracture Mechanics

Certificates

Machine Learning — Stanford University, Jan 2018

Projects

Quantum Computing — Jan 2018 – Jan 2018

Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Jung Heon Lee, PhD

Jung Heon Lee is a professor at Sungkyunkwan University. He is a leading expert in the field of materials science and engineering, with a particular focus on the development of biomaterials for use in a variety of applications.